

Corrosion

Corrosion is an undesirable process. Due to corrosion there is limitation of progress in many areas. The cost of replacement of materials and equipments lost through corrosion is unlimited. Metals and alloys are used as fabrication or construction materials in engineering. If the metals or alloy structures are not properly maintained, they deteriorate slowly by the action of atmospheric gases, moisture and other chemicals. This phenomenon of destruction of metals and alloys is known as corrosion.

Corrosion of metals is defined as the spontaneous destruction of metals in the course of their chemical, electrochemical or biochemical interactions with the environment. Thus, it is exactly the reverse of extraction of metals from ores.
Eg: (i) Rusting of iron (A layer of reddish scale and powder of oxide (Fe_3O_4) is formed on the surface of iron metal.

(ii) A green film of basic carbonate [$\text{CuCO}_3 + \text{Cu}(\text{OH})_2$] is formed on the surface of copper, when it is exposed to moist-air containing carbon dioxide.

CONSEQUENCES (EFFECTS) OF CORROSION:

The economic and social consequences of corrosion include

- i) Due to formation of corrosion product over the machinery, the efficiency of the machine gets failure leads to plant shut down.
- ii) The products contamination or loss of products due to corrosion.
- iii) The corroded equipment must be replaced
- iv) Preventive maintenance like metallic coating or organic coating is required.
- v) Corrosion releases the toxic products.
- vi) Health (eg., from pollution due to a corrosion product or due to the escaping chemical from a corroded equipment).

CAUSES OF CORROSION:

In nature, metals occur in two different forms.

- 1) Native State
- (2) Combined State

Example: (i) Silver materials undergo chemical corrosion by Atmospheric H_2S gas .
(ii) Iron metal undergo chemical corrosion by HCl gas.

CORROSION BY OXYGEN or OXIDATION CORROSION:

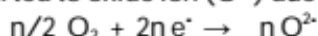
Oxidation Corrosion is brought about by the direct attack of oxygen at low or high temperature on metal surfaces in the absence of moisture. Alkali metals (Li, Na, K etc.,) and alkaline earth metals (Mg, Ca, Sn, etc.,) are rapidly oxidized at low temperature. At high temperature, almost all metals (except Ag, Au and Pt) are oxidized. The reactions of oxidation corrosion are as follows:

Mechanism:

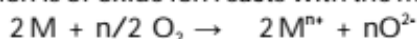
- 1) Oxidation takes place at the surface of the metal forming metal ions M^{2+}



- 2) Oxygen is converted to oxide ion (O^{2-}) due to the transfer of electrons from metal.



- 3) The overall reaction is of oxide ion reacts with the metal ions to form metal oxide film.



The Nature of the Oxide formed plays an important part in oxidation corrosion process.



When oxidation starts, a thin layer of oxide is formed on the metal surface and the nature of this film decides the further action. If the film is

Corrosion by other gases (by hydrogen):

1) Hydrogen Embrittlement:

Loss in ductility of a material in the presence of hydrogen is known as hydrogen embrittlement.

Mechanism:

This type of corrosion occurs when a metal is exposed to hydrogen environment. Iron liberates atomic hydrogen with hydrogen sulphide in the following way.



Hydrogen diffuses into the metal matrix in this atomic form and gets collected in the voids present inside the metal. Further, diffusion of atomic hydrogen makes them combine with each other and forms hydrogen gas.



Collection of these gases in the voids develops very high pressure, causing cracking or blistering of metal.

2) Decarburisation:

The presence of carbon in steel gives sufficient strength to it. But when steel is exposed to hydrogen environment at high temperature, atomic hydrogen is formed.



Atomic hydrogen reacts with the carbon of the steel and produces methane gas.



Hence, the carbon content in steel is decreases. The process of decrease in carbon content in steel is known as decarburization.

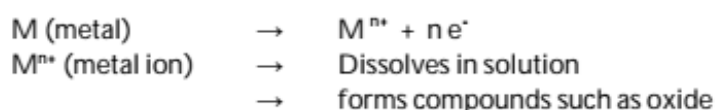
Collection of methane gas in the voids of steel develops high pressure, which causes cracking. Thus, steel loses its strength.

WET OR ELECTROCHEMICAL CORROSION

Electrochemical corrosion involves:

- i) The formation of anodic and cathodic areas or parts in contact with each other
- ii) Presence of a conducting medium
- iii) Corrosion of anodic areas only and
- iv) Formation of corrosion product somewhere between anodic and cathodic areas. This involves flow of electron-current between the anodic and cathodic areas.

At anodic area oxidation reaction takes place (liberation of free electron), so anodic metal is destroyed by either dissolving or assuming combined state (such as oxide, etc.). Hence corrosion always occurs at anodic areas.



At cathodic area, reduction reaction takes place (gain of electrons), usually cathode reactions do not affect the cathode, since most metals cannot be further reduced. So at cathodic part, dissolved constituents in the conducting medium accepts the electrons to form some ions like OH^- and O_2^- .

Cathodic reaction consumes electrons with either by

- (a) evolution of hydrogen or
- (b) absorption of oxygen, depending on the nature of the corrosive environment

Hydrogen Evolution Type:

All metals above hydrogen in the electrochemical series have a tendency to get dissolved in acidic solution with simultaneous evolution of hydrogen.

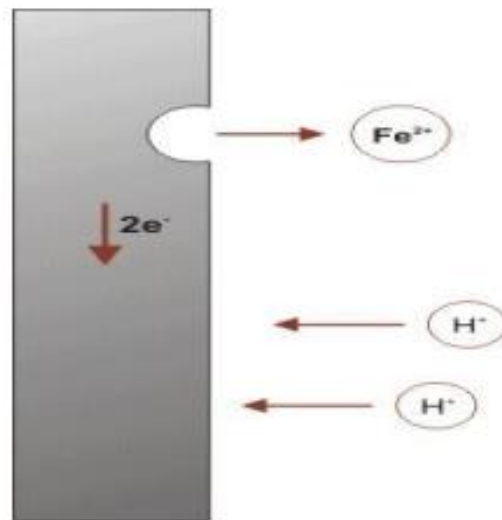
It occurs in acidic environment. Consider the example of iron

At anode: $\text{Fe} \rightarrow \text{Fe}^{2+} + 2\text{e}^-$

These electrons flow through the metal, from anode to cathode, where H^+ ions of acidic solution are eliminated as hydrogen gas.

At cathode: $2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2\uparrow$

The overall reaction is: $\text{Fe} + 2\text{H}^+ \rightarrow \text{Fe}^{2+} + \text{H}_2$



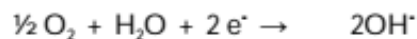
Oxygen Absorption Type:

Rusting of iron in neutral aqueous solution of electrolytes (like NaCl solution) in the presence of atmospheric oxygen is a common example of this type of corrosion. The surface of iron is usually coated with a thin film of iron oxide. However, if this iron oxide film develops some cracks, anodic areas are created on the surface; while the well metal parts acts as cathodes.

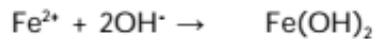
At Anode: Metal dissolves as ferrous ions with liberation of electrons.



At Cathode: The liberated electrons are intercepted by the dissolved oxygen.



The Fe^{2+} ions and OH^{-} ions diffuse and when they meet, ferrous hydroxide is precipitated.



(i) If enough oxygen is present, ferrous hydroxide is easily oxidized to ferric hydroxide.



(ii) If the supply of oxygen is limited, the corrosion product may be even black anhydrous magnetite, Fe_3O_4 .